

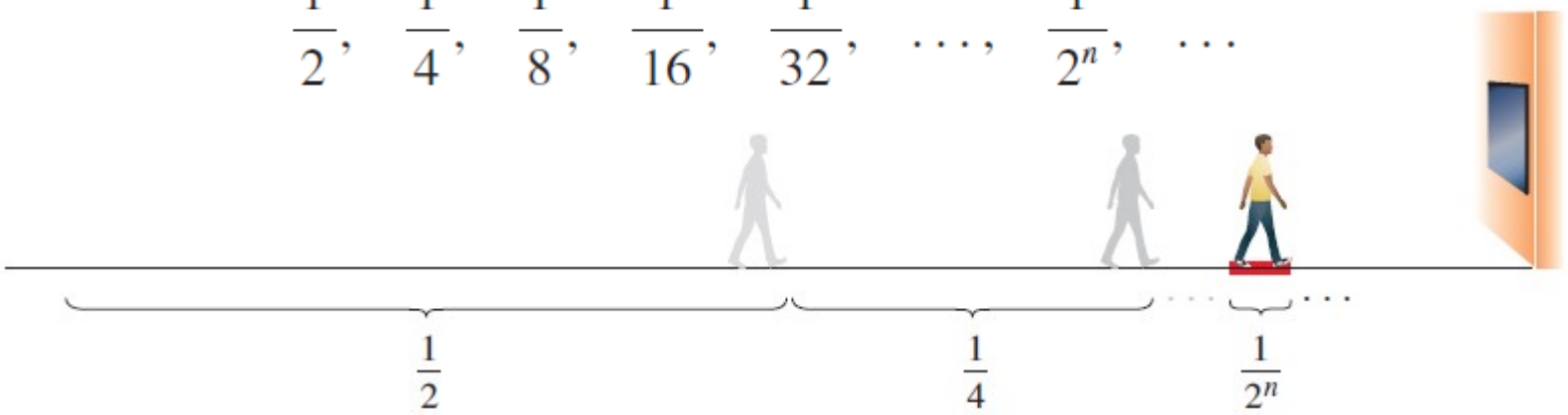
Chapter 9:

Infinite Series

Section 9.2: Series and Convergence

Zeno's Paradox (Dichotomy Paradox)

$$\frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \frac{1}{16}, \quad \frac{1}{32}, \quad \cdots, \quad \frac{1}{2^n}, \quad \cdots$$



$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} + \cdots$$

There are situations in which we implicitly use infinite sums. For instance, in decimal notation, the value of π is

$$\pi = 3.14159\ 26535\ 89793\ 23846\ 26433\ 83279\ 50288\ \ldots$$

The convention behind our decimal notation is that this number can be written as the infinite sum

$$\pi = 3 + \frac{1}{10} + \frac{4}{10^2} + \frac{1}{10^3} + \frac{5}{10^4} + \frac{9}{10^5} + \frac{2}{10^6} + \frac{6}{10^7} + \frac{5}{10^8} + \cdots$$

Infinite Series

If we try to add the terms of an infinite sequence $\{a_n\}_{n=1}^{\infty}$ we get an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots \quad (1)$$

which is called an **infinite series** (or just a **series**) and is denoted by the symbol

$$\sum_{n=1}^{\infty} a_n \qquad \text{or} \qquad \sum a_n$$

Definition: Let $\sum_{n=1}^{\infty} a_n$ be a series. The sum of the first n terms is called the ***n*th partial sum of the series** and we denote it by s_n.

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$s_3 = a_1 + a_2 + a_3$$
$$s_4 = a_1 + a_2 + a_3 + a_4$$

and, in general, $$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{i=1}^{n} a_i$$

EXAMPLE

Consider the series of distances from Zeno's paradox:

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\cdots+\frac{1}{2^n}+\cdots$$

If we start adding the terms, and we keep track of the subtotals as we go, we get $\frac{1}{2}$, $\frac{3}{4}$ (the sum of the first two terms), $\frac{7}{8}$ (first three terms), $\frac{15}{16}$, $\frac{31}{32}$, $\frac{63}{64}$, and so on. The table in the margin shows that as we add more and more terms, these *partial sums* become closer and closer to 1.

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

So it seems reasonable to say that the sum of this infinite series is 1 and to write

$$\sum_{n=1}^{\infty}\frac{1}{2^n}=\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots+\frac{1}{2^n}+\cdots=1$$

Definitions of Convergent and Divergent Series

For the infinite series $\sum_{n=1}^{\infty} a_n$, the ***n*th partial sum** is

$$S_n = a_1 + a_2 + \cdots + a_n.$$

If the sequence of partial sums $\{S_n\}$ converges to S, then the series $\sum_{n=1}^{\infty} a_n$ **converges.** The limit S is called the **sum of the series.**

$$S = a_1 + a_2 + \cdots + a_n + \cdots \qquad S = \sum_{n=1}^{\infty} a_n$$

If $\{S_n\}$ diverges, then the series **diverges.**

Notice that

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} \sum_{i=1}^{n} a_i$$

EXAMPLE

Suppose we know that the sum of the first n terms of the series $\sum_{n=1}^{\infty} a_n$ is

$$s_n = a_1 + a_2 + \cdots + a_n = \frac{2n}{3n + 5}$$

Then the sum of the series is the limit of the sequence $\{s_n\}$:

$$\sum_{n=1}^{\infty} a_n = \lim_{n \to \infty} s_n = \lim_{n \to \infty} \frac{2n}{3n + 5} = \lim_{n \to \infty} \frac{2}{3 + \frac{5}{n}} = \frac{2}{3}$$

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EXAMPLE Telescoping Series

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

SOLUTION

EXAMPLE

Telescoping Series

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$$

SOLUTION

The series in slides 7 and 9 are called **telescoping series** of the form

$$(b_1 - b_2) + (b_2 - b_3) + (b_3 - b_4) + (b_4 - b_5) + \cdots.$$

Telescoping series

Note that b_2 is canceled by the second term, b_3 is canceled by the third term, and so on. Because the nth partial sum of this series is

$$S_n = b_1 - b_{n+1}$$

it follows that a telescoping series will converge if and only if b_n approaches a finite number as $n \to \infty$. Moreover, if the series converges, then its sum is

$$S = b_1 - \lim_{n \to \infty} b_{n+1}.$$

Sum of a Geometric Series

An important example of infinite series is the **geometric series**

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots + ar^n + \cdots, \quad a \neq 0$$

Geometric series

Each term is obtained from the preceding one by multiplying by the **common ratio** r.
Notice that a is the **first term** of the geometric series.

THEOREM 9.6 Convergence of a Geometric Series

A geometric series with ratio r diverges when $|r| \geq 1$. If $|r| < 1$, then the series converges to the sum

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad |r| < 1.$$



EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

33. $8 + 6 + \frac{9}{2} + \frac{27}{8} + \cdots$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=0}^{\infty} 5\left(-\frac{3}{2}\right)^n$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=0}^{\infty} 2^{2n}\, 3^{1-n}$$

SOLUTION

EXAMPLE

Write the number $2.3\overline{17} = 2.3171717\ldots$ as a ratio of integers.

SOLUTION

$$2.3171717\ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \cdots$$

After the first term we have a geometric series with $a = 17/10^3$ and $r = 1/10^2$. Therefore

$$2.3\overline{17} = 2.3 + \frac{\dfrac{17}{10^3}}{1 - \dfrac{1}{10^2}} = 2.3 + \frac{\dfrac{17}{1000}}{\dfrac{99}{100}}$$

$$= \frac{23}{10} + \frac{17}{990} = \frac{1147}{495}$$

The convergence of a series is not affected by the removal of a finite number of terms from the beginning of the series. For instance, the geometric series

$$\sum_{n=4}^{\infty} \left(\frac{1}{2}\right)^n \quad \text{and} \quad \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

both converge. Furthermore, because the sum of the second series is

$$\frac{a}{1-r} = \frac{1}{1-(1/2)} = 2$$

you can conclude that the sum of the first series is

$$\begin{aligned} S &= 2 - \left[\left(\frac{1}{2}\right)^0 + \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3\right] \\ &= 2 - \frac{15}{8} \\ &= \frac{1}{8}. \end{aligned}$$

THEOREM 9.7 Properties of Infinite Series

Let Σa_n and Σb_n be convergent series, and let A, B, and c be real numbers. If $\Sigma a_n = A$ and $\Sigma b_n = B$, then the following series converge to the indicated sums.

1. $\displaystyle\sum_{n=1}^{\infty} ca_n = cA$
2. $\displaystyle\sum_{n=1}^{\infty} (a_n + b_n) = A + B$
3. $\displaystyle\sum_{n=1}^{\infty} (a_n - b_n) = A - B$

THEOREM 9.8 Limit of the *n*th Term of a Convergent Series

If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Proof Assume that

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = L.$$

Then, because $S_n = S_{n-1} + a_n$ and

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} S_{n-1} = L$$

it follows that

$$\begin{aligned} L &= \lim_{n\to\infty} S_n \\ &= \lim_{n\to\infty} (S_{n-1} + a_n) \\ &= \lim_{n\to\infty} S_{n-1} + \lim_{n\to\infty} a_n \\ &= L + \lim_{n\to\infty} a_n \end{aligned}$$

which implies that $\{a_n\}$ converges to 0. ∎

THEOREM 9.8 Limit of the *n*th Term of a Convergent Series

If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

The contrapositive of Theorem 9.8 provides a useful test for *divergence*. This ***n*th-Term Test for Divergence** states that if the limit of the nth term of a series does *not* converge to 0, then the series must diverge.

THEOREM 9.9 *n*th-Term Test for Divergence

If $\lim_{n\to\infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ diverges.

WARNING: The converse of Theorem 9.9 is not true in general. That is, if $\lim_{n\to\infty} a_n = 0$, we cannot conclude that the series $\sum a_n$ is convergent.

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty}(-1)^n$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} n$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} \frac{-n}{3n+1}$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} \tan^{-1} n$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$$

SOLUTION

Properties of Convergent Series

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=1}^{\infty}\left[\frac{4}{n(n+1)}+\left(\frac{3}{4}\right)^{n}\right]$$

SOLUTION

EXAMPLE

Determine whether the series is convergent or divergent. If it is convergent find its sum.

$$\sum_{n=0}^{\infty} \frac{e^{1-2n}}{\left(\sqrt{2}\right)^{2-2n}}$$

SOLUTION

The series is convergent, and its sum is $s = \frac{e^3}{2(e^2 - 2)}$.

NOTE A finite number of terms doesn't affect the convergence or divergence of a series. For instance, suppose that we were able to show that the series

$$\sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

is convergent. Since

$$\sum_{n=1}^{\infty} \frac{n}{n^3 + 1} = \frac{1}{2} + \frac{2}{9} + \frac{3}{28} + \sum_{n=4}^{\infty} \frac{n}{n^3 + 1}$$

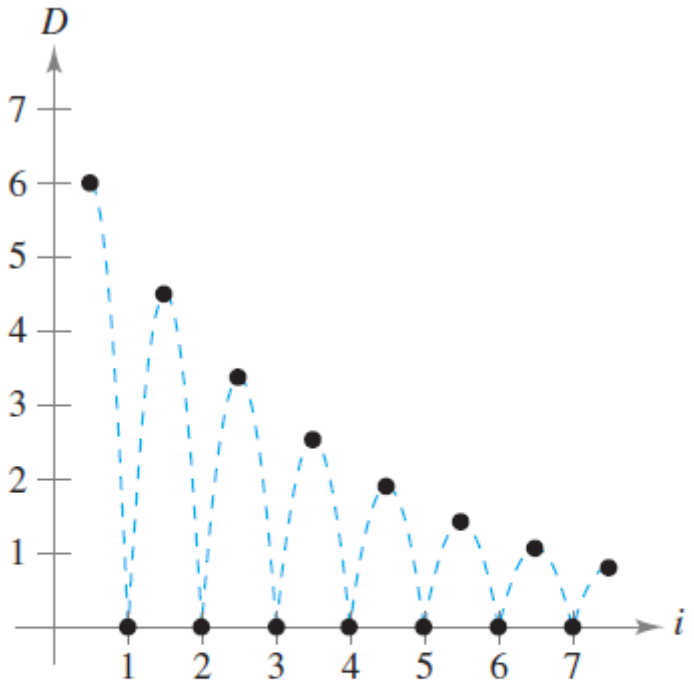
it follows that the entire series $\sum_{n=1}^{\infty} n/(n^3 + 1)$ is convergent. Similarly, if it is known that the series $\sum_{n=N+1}^{\infty} a_n$ converges, then the full series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{N} a_n + \sum_{n=N+1}^{\infty} a_n$$

is also convergent.

EXAMPLE 6 Bouncing Ball Problem

A ball is dropped from a height of 6 feet and begins bouncing, as shown in Figure 9.7. The height of each bounce is three-fourths the height of the previous bounce. Find the total vertical distance traveled by the ball.



The height of each bounce is three-fourths the height of the preceding bounce.

Figure 9.7

Solution When the ball hits the ground for the first time, it has traveled a distance of $D_1 = 6$ feet. For subsequent bounces, let D_i be the distance traveled up and down. For example, D_2 and D_3 are

$$D_2 = \underbrace{6\left(\frac{3}{4}\right)}_{\text{Up}} + \underbrace{6\left(\frac{3}{4}\right)}_{\text{Down}} = 12\left(\frac{3}{4}\right)$$

and

$$D_3 = \underbrace{6\left(\frac{3}{4}\right)\left(\frac{3}{4}\right)}_{\text{Up}} + \underbrace{6\left(\frac{3}{4}\right)\left(\frac{3}{4}\right)}_{\text{Down}} = 12\left(\frac{3}{4}\right)^2.$$

By continuing this process, it can be determined that the total vertical distance is

$$\begin{aligned} D &= 6 + 12\left(\frac{3}{4}\right) + 12\left(\frac{3}{4}\right)^2 + 12\left(\frac{3}{4}\right)^3 + \cdots \\ &= 6 + 12\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^{n+1} \\ &= 6 + 12\left(\frac{3}{4}\right)\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^{n} \\ &= 6 + 9\left[\frac{1}{1-(3/4)}\right] \\ &= 6 + 9(4) \\ &= 42 \text{ feet.} \end{aligned}$$

Solve exercises of section 9.2.

Start with the suggested problems mentioned in the syllabus.